

Algebra 2

Unit 6: Solving Cube Root Equations

Day 2: Complex Equations & Applications

I can solve multistep cube root equations.

Name: _____

Quick Review: Steps for Solving

1. Isolate the radical or power (get the expression with the exponent by itself)
2. Cube both sides (or raise to the reciprocal power) to eliminate the radical
3. Solve for x

Part 1: Multi-Step Isolating

1. $4\sqrt[3]{x+2} + 5 = -3$

3. $\frac{2}{3}(x-5)^{\frac{1}{3}} - 4 = -2$

2. $5(x-2)^{\frac{2}{3}} + 7 = 52$

4. $3 + 2(x-1)^{\frac{1}{3}} = 7$

Part 2: Coefficients on Both Sides

1. $3\sqrt[3]{x+5} = 6\sqrt[3]{x+4}$

2. $2\sqrt[3]{10x+1} = 3\sqrt[3]{3x-1}$

Part 3: Applications

1. A shipping container is a perfect cube with volume 729 cubic feet. What is the side length s ?

2. Wind power $P = 0.2v^3$. If the turbine generates 54 units of power, find the wind velocity v .

You Try!

1. $3\sqrt[3]{2x+1} + 5 = 14$

6. $2\sqrt[3]{x+3} + 1 = 6$

2. $4\sqrt[3]{x+20} - 2 = -6$

7. $-7 + 4\sqrt[3]{x+4} = 5$

3. $5(x-3)^{\frac{1}{3}} = 12$

8. $\frac{1}{2}(4x+4)^{\frac{1}{3}} + 3 = 1$

4. $2(3x+5)^{\frac{2}{3}} - 4 = 14$

9. $5(3x-2)^{\frac{2}{3}} - 8 = 12$

5. A cube has volume 64 ft^3 . Find s .

10. The surface area of a cube is $SA = 6s^2$. If the volume $V = 8 \text{ cm}^3$, find the surface area.

Answer Bank

-21
-19
-15
-10
-3
1
2
3
4
6
9
11
13
23
24
25
 $\frac{101}{8}$
 $\frac{2103}{125}$